

MODULE 07: Master Regression — Sentiment -> CAR Analysis Dataset (merged)

Obs	fyear	CAR_0_3	neg_pct	unc_pct	net_tone	surprise_pct
1	2010	(1.744%)	1.701	1.584	-0.978	4.06%
2	2011	(0.879%)	1.735	1.792	-1.043	0.88%
3	2012	(3.994%)	1.598	1.744	-0.889	(0.20%)
4	2013	1.742%	1.663	1.710	-1.013	1.40%
5	2014	(0.187%)	1.598	1.654	-0.932	2.06%
6	2015	3.564%	1.653	1.728	-0.946	1.10%
7	2016	(4.029%)	1.614	1.763	-0.920	0.36%
8	2017	4.433%	1.621	1.788	-0.922	2.33%
9	2018	(2.881%)	1.745	1.773	-1.026	1.28%
10	2019	3.647%	1.887	1.635	-1.277	1.80%
11	2020	(5.506%)	1.876	1.600	-1.319	1.23%
12	2021	(2.786%)	2.102	1.529	-1.486	0.18%
13	2022	1.804%	2.095	1.570	-1.493	0.16%
14	2023	1.764%	2.183	1.572	-1.590	1.16%
15	2024	(5.999%)	2.325	1.551	-1.673	1.20%

MODULE 07: Master Regression — Sentiment -> CAR

Descriptive Statistics

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
CAR_0_1	15	-0.0049	0.0313	-0.0594	0.0408
CAR_0_3	15	-0.0074	0.0344	-0.0600	0.0443
CAR_0_5	15	-0.0135	0.0419	-0.0702	0.0469
CAR_1_10	15	-0.0248	0.0326	-0.0709	0.0418
neg_pct	15	1.8266	0.2403	1.5983	2.3246
pos_pct	15	0.6595	0.0526	0.5576	0.7233
unc_pct	15	1.6662	0.0947	1.5293	1.7916
net_tone	15	-1.1670	0.2772	-1.6734	-0.8885
surprise_pct	15	0.0127	0.0105	-0.0020	0.0406
sue_stdev	15	1.4103	1.4761	-0.1600	5.2500

MODULE 07: Master Regression — Sentiment -> CAR

Correlations: CARs vs Sentiment and Surprise

The CORR Procedure

4 With Variables:	neg_pct	unc_pct	net_tone	surprise_pct
4 Variables:	CAR_0_1	CAR_0_3	CAR_0_5	CAR_1_10

Simple Statistics						
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
neg_pct	15	1.82656	0.24034	27.39844	1.59828	2.32460
unc_pct	15	1.66623	0.09473	24.99339	1.52926	1.79157
net_tone	15	-1.16705	0.27717	-17.50572	-1.67344	-0.88853
surprise_pct	15	0.01266	0.01050	0.18987	-0.00203	0.04058
CAR_0_1	15	-0.00492	0.03134	-0.07380	-0.05937	0.04083
CAR_0_3	15	-0.00737	0.03436	-0.11050	-0.05999	0.04433
CAR_0_5	15	-0.01345	0.04191	-0.20182	-0.07017	0.04685
CAR_1_10	15	-0.02476	0.03259	-0.37146	-0.07087	0.04177

Pearson Correlation Coefficients, N = 15 Prob > r under H0: Rho=0				
	CAR_0_1	CAR_0_3	CAR_0_5	CAR_1_10
neg_pct	-0.06658 0.8136	-0.18745 0.5035	-0.14199 0.6137	0.14732 0.6003
unc_pct	0.04822 0.8645	0.18588 0.5071	0.03861 0.8913	-0.22363 0.4230
net_tone	0.06354 0.8220	0.16290 0.5619	0.11412 0.6855	-0.17065 0.5431
surprise_pct	0.19501 0.4861	0.24789 0.3730	0.48185 0.0689	0.49445 0.0610

MODULE 07: Master Regression — Sentiment -> CAR
MAIN REGRESSION: CAR[0,+3] = f(Negative Sentiment, Earnings Surprise)

The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_0_3

Number of Observations Read	15
Number of Observations Used	15

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	0.00131	0.00065568	0.52	0.6090
Error	12	0.01522	0.00127		
Corrected Total	14	0.01653			

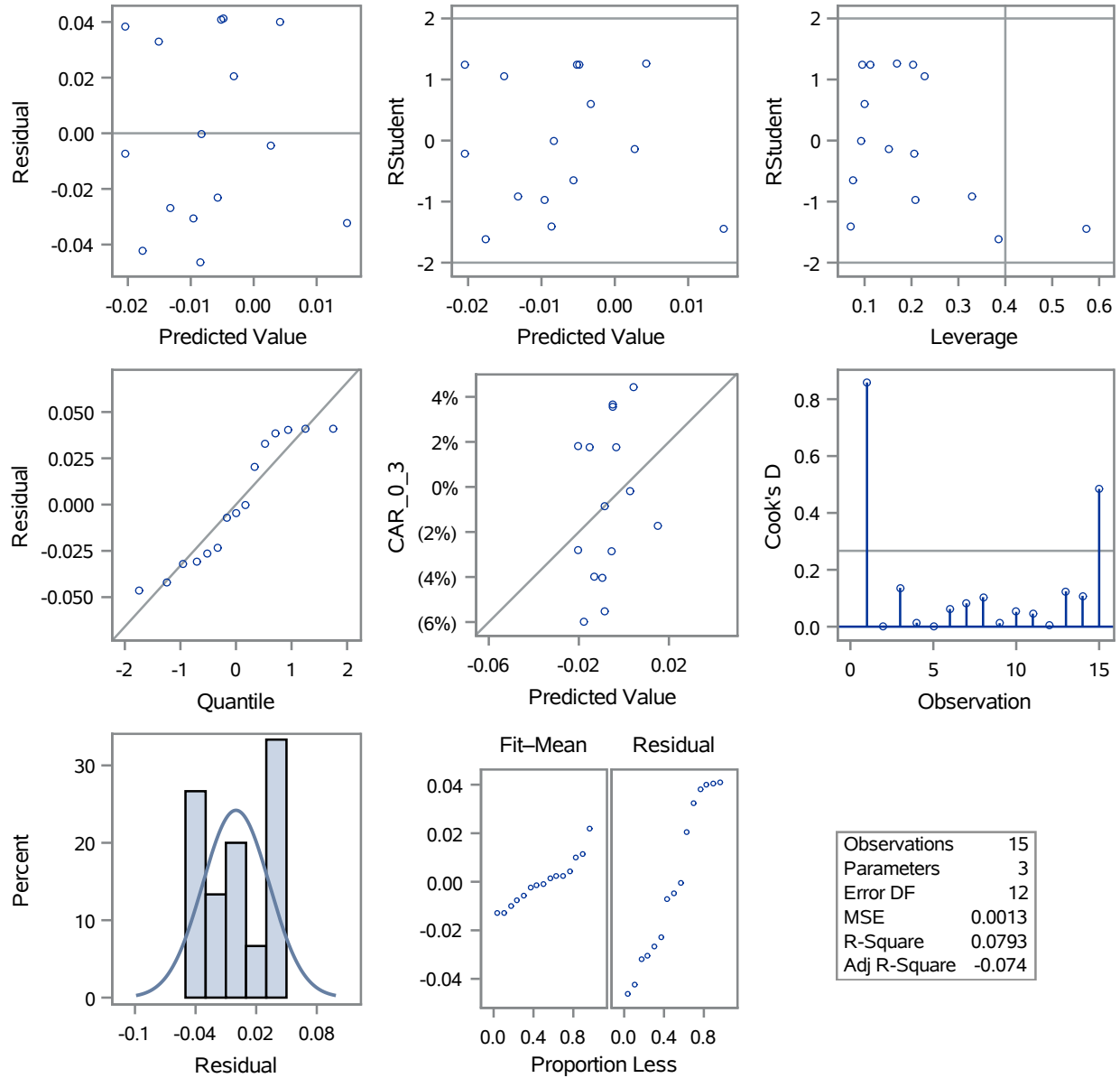
Root MSE	0.03561	R-Square	0.0793
Dependent Mean	-0.00737	Adj R-Sq	-0.0741
Coeff Var	-483.44003		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	0.01957	0.07848	0.25	0.8073
neg_pct	1	-0.01965	0.04071	-0.48	0.6380
surprise_pct	1	0.70733	0.93206	0.76	0.4626

MODULE 07: Master Regression — Sentiment -> CAR
MAIN REGRESSION: CAR[0,+3] = f(Negative Sentiment, Earnings Surprise)

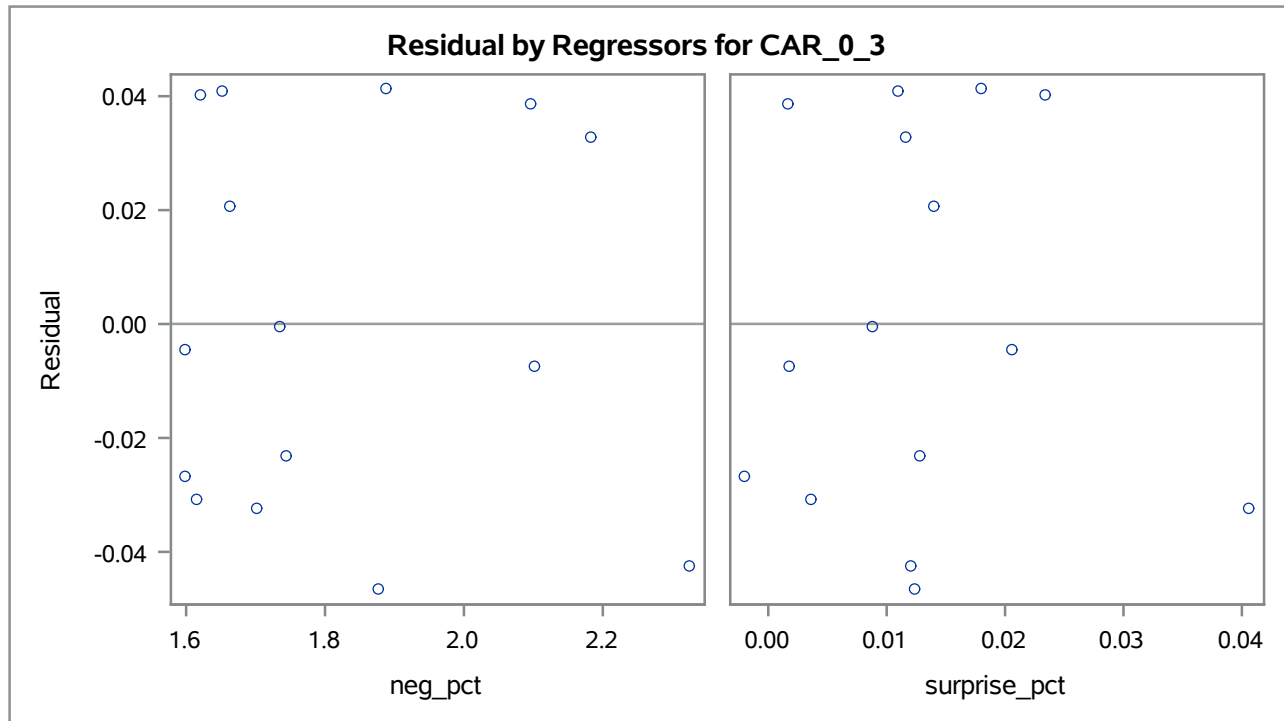
The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_0_3

Fit Diagnostics for CAR_0_3



MODULE 07: Master Regression — Sentiment -> CAR
MAIN REGRESSION: CAR[0,+3] = f(Negative Sentiment, Earnings Surprise)

The REG Procedure
Model: MODEL1
Dependent Variable: CAR_0_3



MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 1: Net Tone instead of Negative Sentiment

The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_0_3

Number of Observations Read	15
Number of Observations Used	15

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	0.00120	0.00059839	0.47	0.6371
Error	12	0.01533	0.00128		
Corrected Total	14	0.01653			

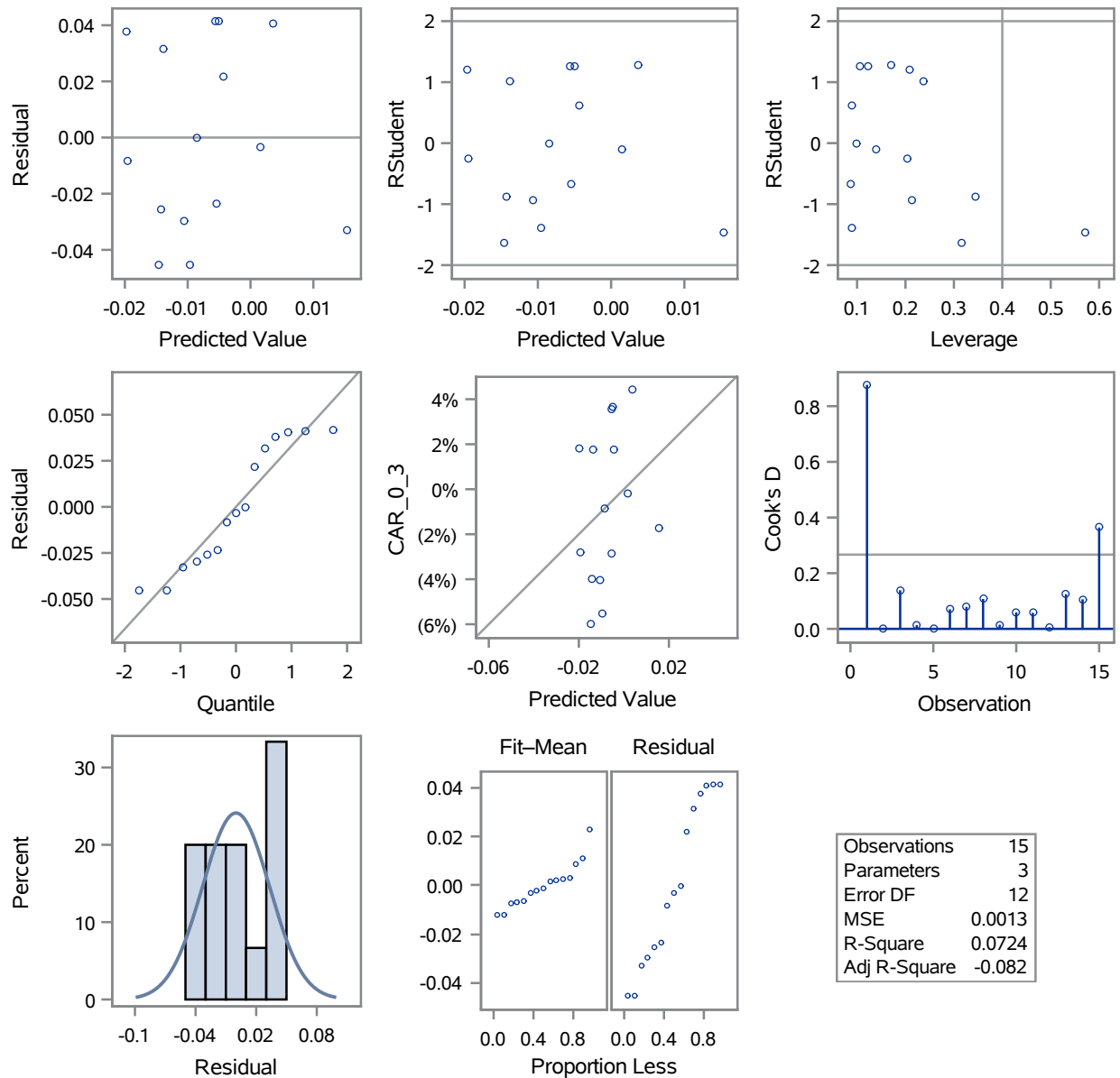
Root MSE	0.03575	R-Square	0.0724
Dependent Mean	-0.00737	Adj R-Sq	-0.0822
Coeff Var	-485.25628		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-0.00090125	0.04687	-0.02	0.9850
net_tone	1	0.01339	0.03559	0.38	0.7133
surprise_pct	1	0.72368	0.93958	0.77	0.4561

MODULE 07: Master Regression — Sentiment -> CAR **ROBUSTNESS 1: Net Tone instead of Negative Sentiment**

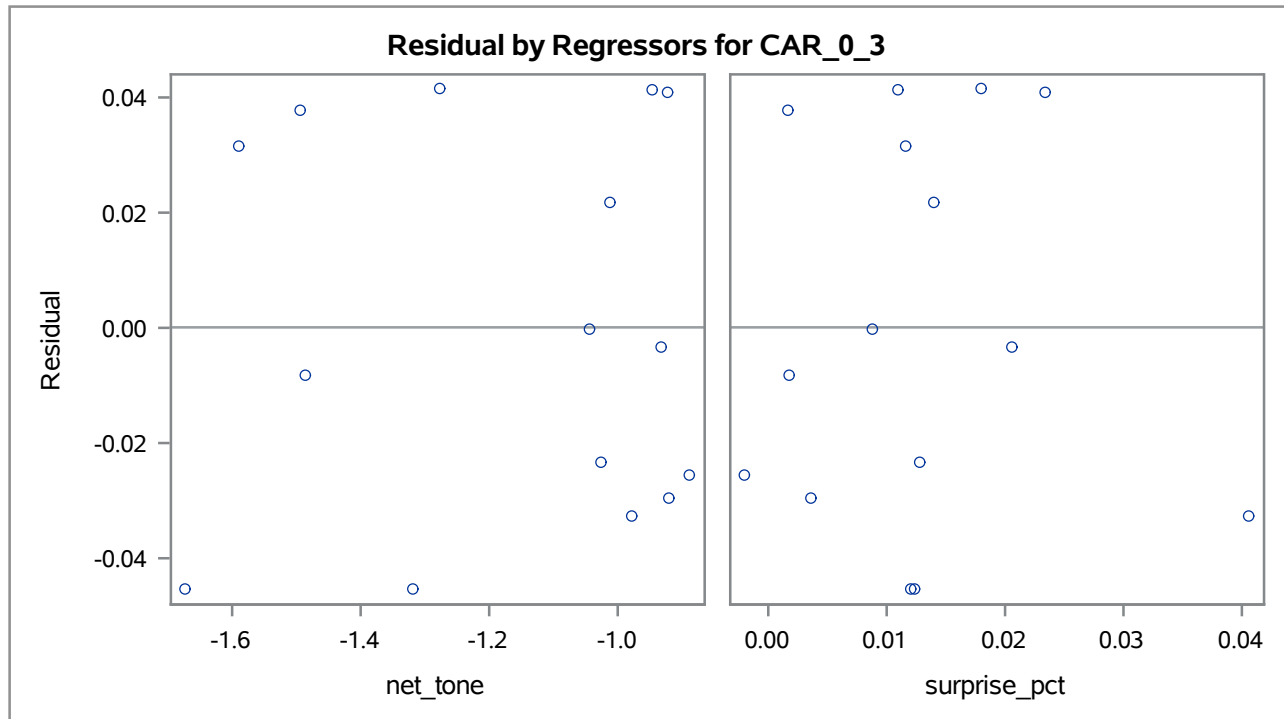
The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_0_3

Fit Diagnostics for CAR_0_3



MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 1: Net Tone instead of Negative Sentiment

The REG Procedure
Model: MODEL1
Dependent Variable: CAR_0_3



MODULE 07: Master Regression — Sentiment -> CAR

ROBUSTNESS 2: Uncertainty sentiment

The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_0_3

Number of Observations Read	15
Number of Observations Used	15

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	0.00171	0.00085432	0.69	0.5197
Error	12	0.01482	0.00124		
Corrected Total	14	0.01653			

Root MSE	0.03515	R-Square	0.1034
Dependent Mean	-0.00737	Adj R-Sq	-0.0461
Coeff Var	-477.08906		

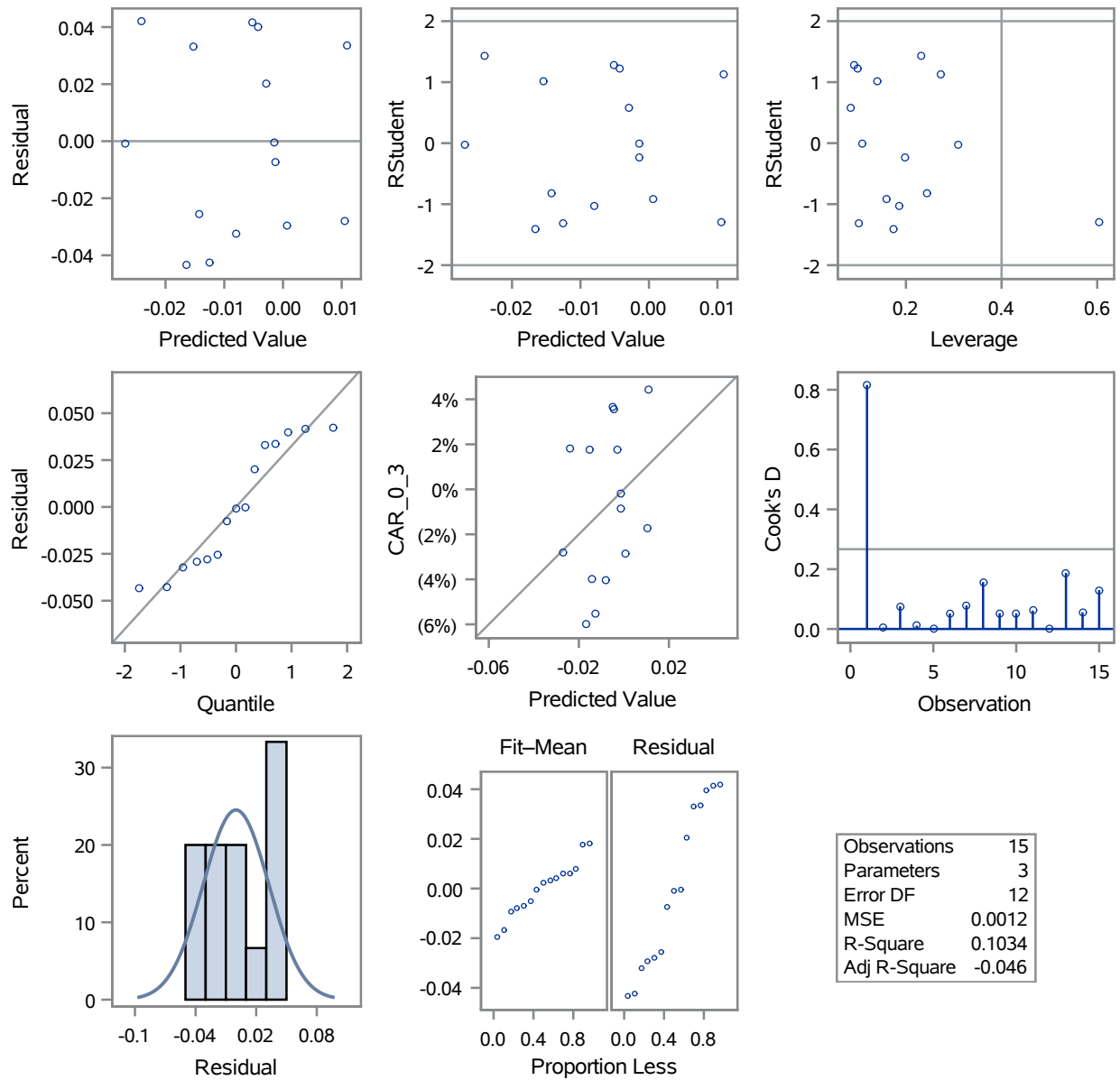
Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-0.14233	0.16713	-0.85	0.4111
unc_pct	1	0.07446	0.09943	0.75	0.4684
surprise_pct	1	0.86100	0.89726	0.96	0.3562

MODULE 07: Master Regression — Sentiment -> CAR

ROBUSTNESS 2: Uncertainty sentiment

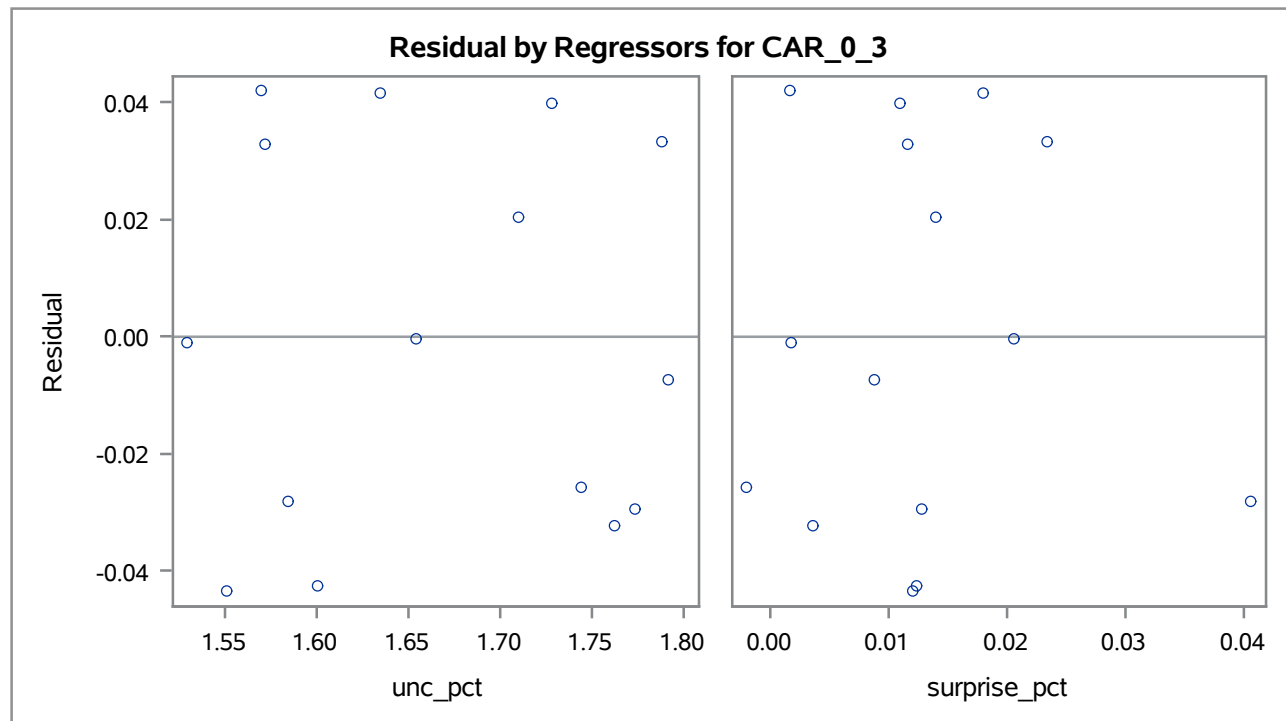
The REG Procedure
Model: MODEL1
Dependent Variable: CAR_0_3

Fit Diagnostics for CAR_0_3



MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 2: Uncertainty sentiment

The REG Procedure
Model: MODEL1
Dependent Variable: CAR_0_3



MODULE 07: Master Regression — Sentiment -> CAR **ROBUSTNESS 3: Shorter window CAR[0,+1]**

The REG Procedure
Model: MODEL1
Dependent Variable: CAR_0_1

Number of Observations Read	15
Number of Observations Used	15

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	0.00052970	0.00026485	0.24	0.7901
Error	12	0.01322	0.00110		
Corrected Total	14	0.01375			

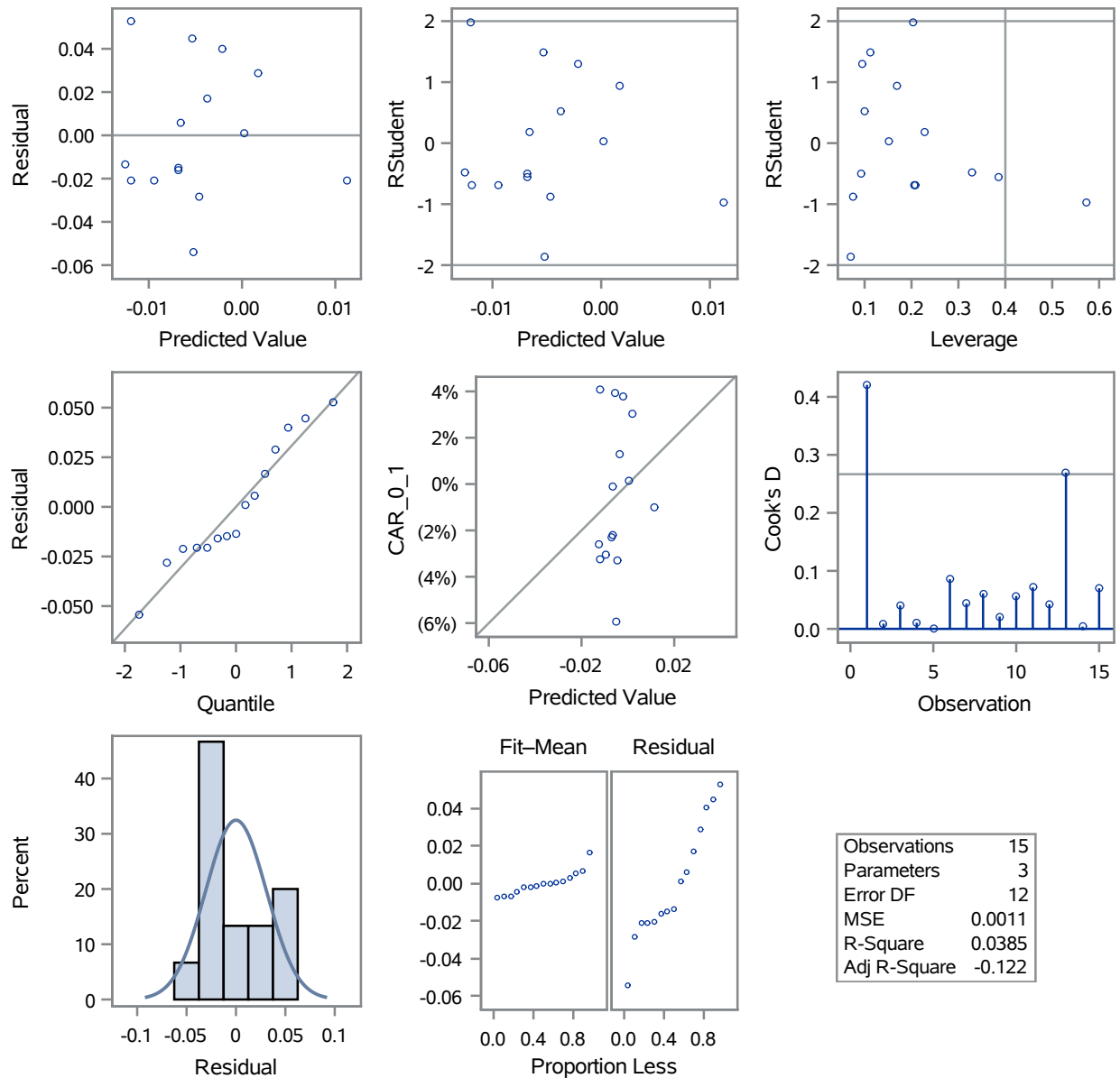
Root MSE	0.03320	R-Square	0.0385
Dependent Mean	-0.00492	Adj R-Sq	-0.1217
Coeff Var	-674.67610		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-0.00670	0.07315	-0.09	0.9286
neg_pct	1	-0.00295	0.03795	-0.08	0.9393
surprise_pct	1	0.56661	0.86878	0.65	0.5266

MODULE 07: Master Regression — Sentiment -> CAR **ROBUSTNESS 3: Shorter window CAR[0,+1]**

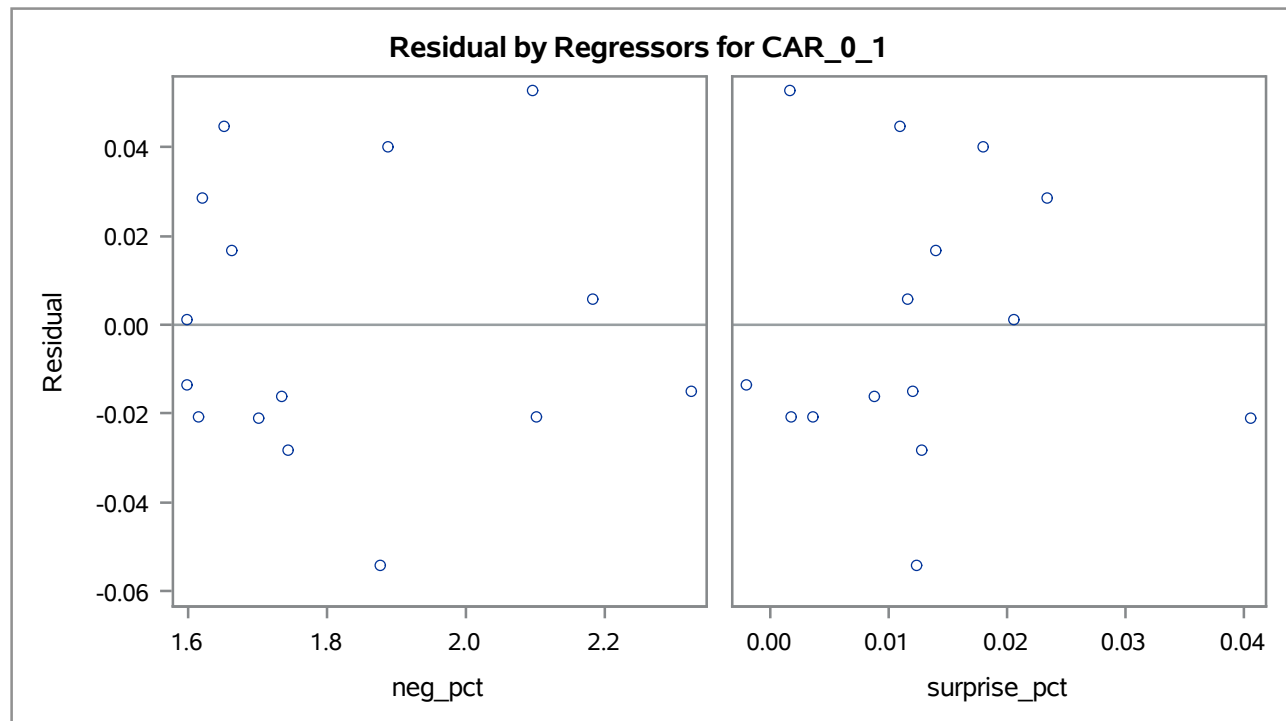
The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_0_1

Fit Diagnostics for CAR_0_1



MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 3: Shorter window CAR[0,+1]

The REG Procedure
Model: MODEL1
Dependent Variable: CAR_0_1



MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 4: Longer window CAR[+1,+10] — post-filing drift

The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_1_10

Number of Observations Read	15
Number of Observations Used	15

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	0.00471	0.00236	2.78	0.1016
Error	12	0.01016	0.00084671		
Corrected Total	14	0.01487			

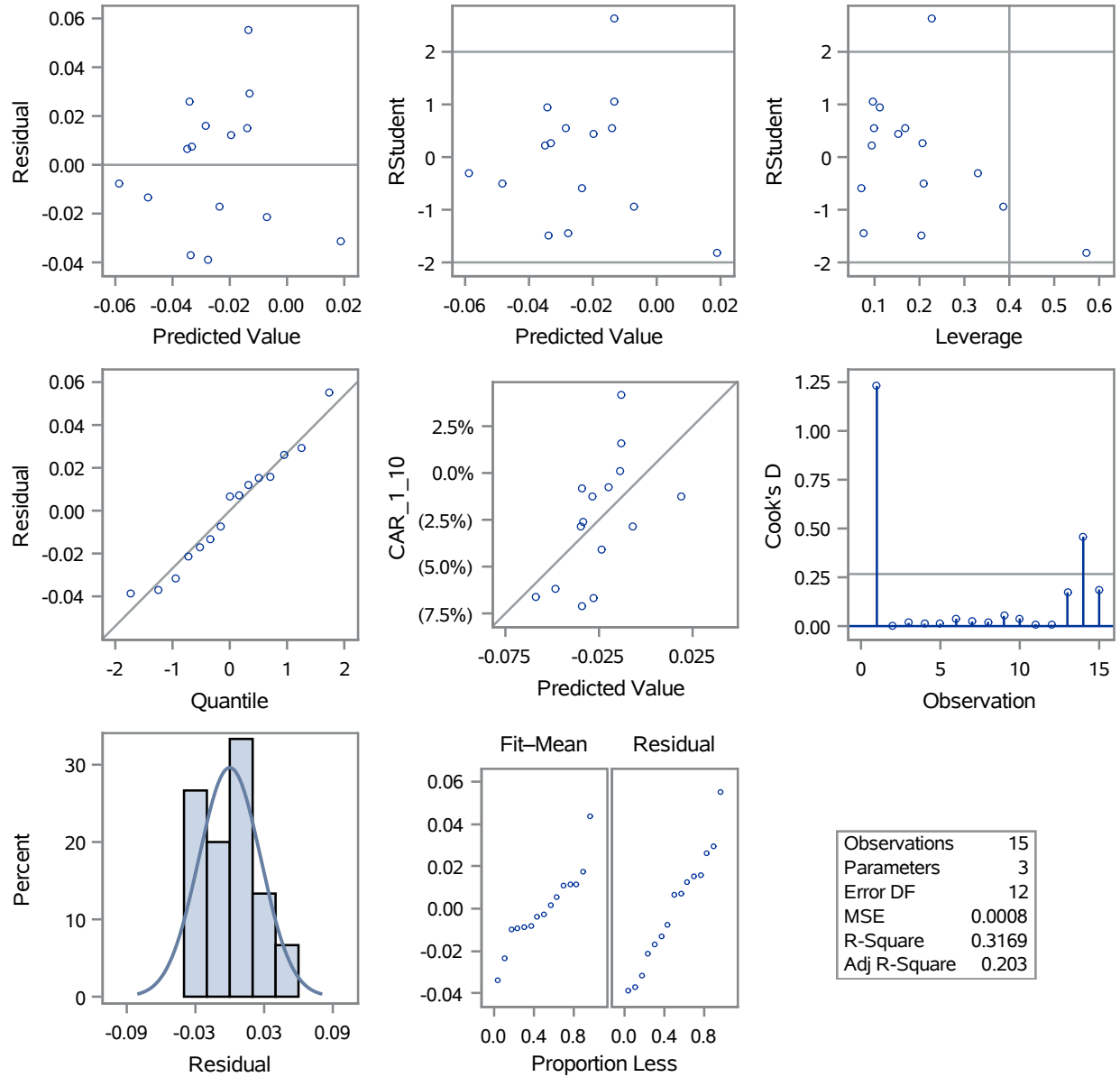
Root MSE	0.02910	R-Square	0.3169
Dependent Mean	-0.02476	Adj R-Sq	0.2030
Coeff Var	-117.50353		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-0.11524	0.06412	-1.80	0.0975
neg_pct	1	0.03752	0.03326	1.13	0.2814
surprise_pct	1	1.73415	0.76153	2.28	0.0419

MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 4: Longer window CAR[+1,+10] — post-filing drift

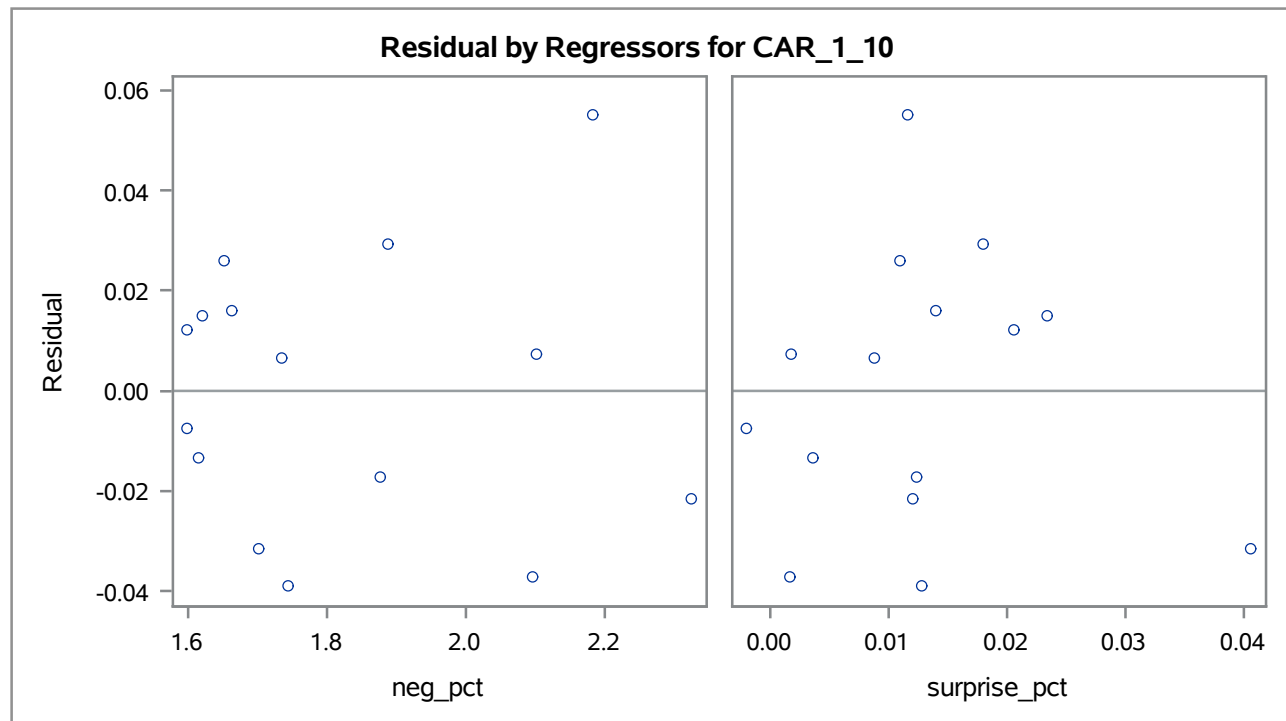
The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_1_10

Fit Diagnostics for CAR_1_10



MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 4: Longer window CAR[+1,+10] — post-filing drift

The REG Procedure
Model: MODEL1
Dependent Variable: CAR_1_10



MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 5: Sentiment alone (no surprise control)

The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_0_3

Number of Observations Read	15
Number of Observations Used	15

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	0.00058090	0.00058090	0.47	0.5035
Error	13	0.01595	0.00123		
Corrected Total	14	0.01653			

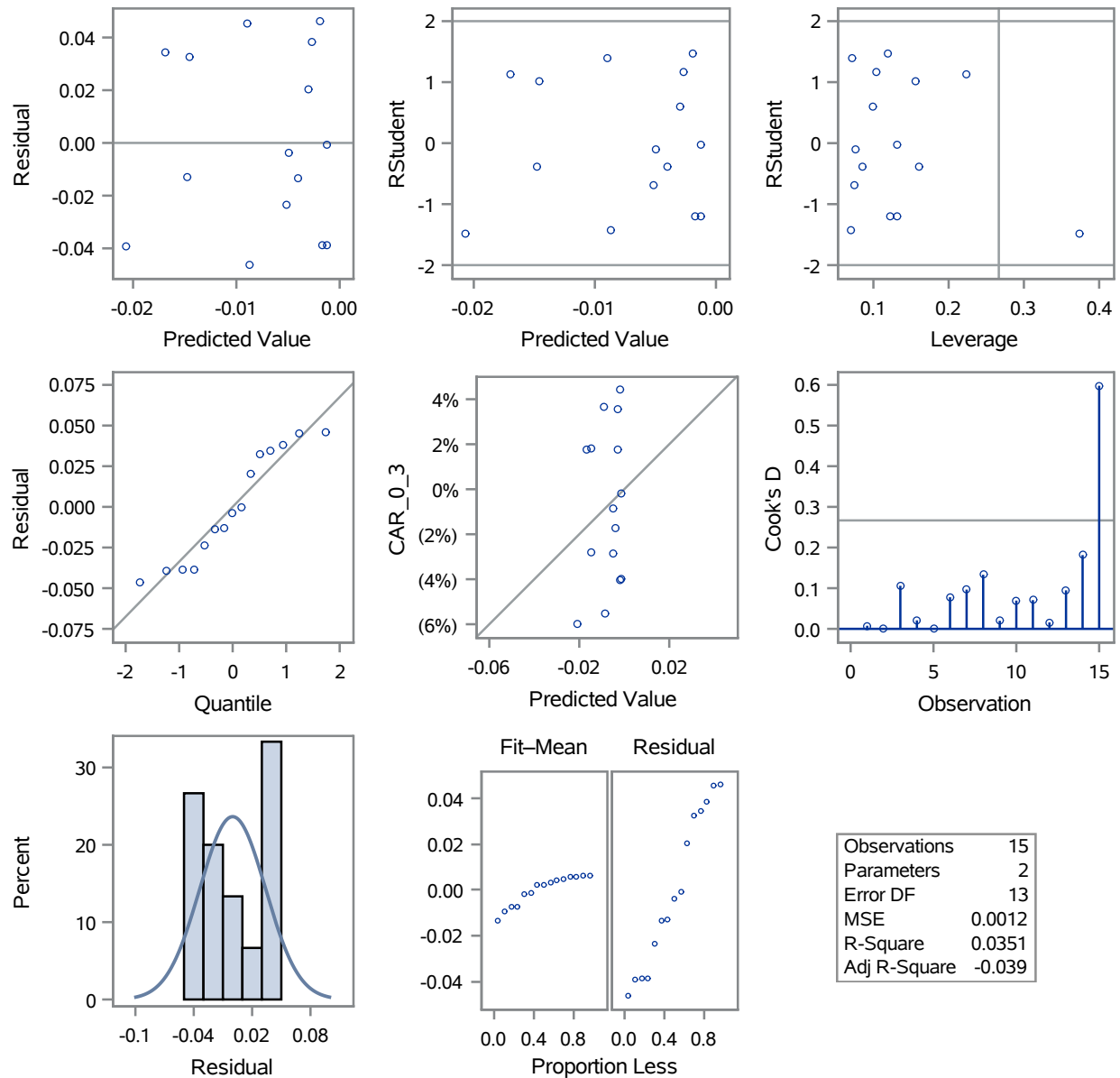
Root MSE	0.03503	R-Square	0.0351
Dependent Mean	-0.00737	Adj R-Sq	-0.0391
Coeff Var	-475.48910		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	0.04159	0.07172	0.58	0.5719
neg_pct	1	-0.02680	0.03895	-0.69	0.5035

MODULE 07: Master Regression — Sentiment -> CAR **ROBUSTNESS 5: Sentiment alone (no surprise control)**

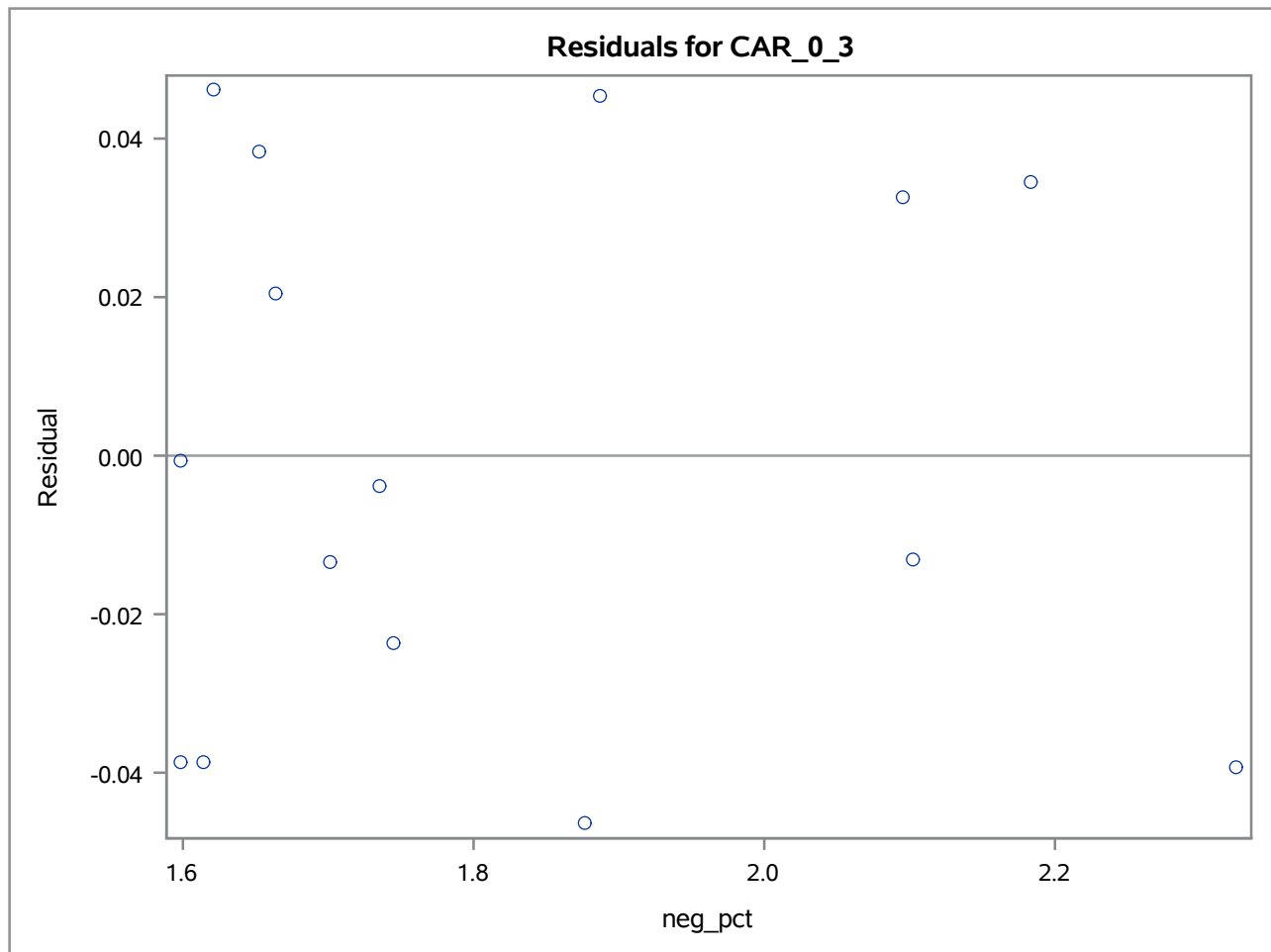
The REG Procedure
 Model: MODEL1
 Dependent Variable: CAR_0_3

Fit Diagnostics for CAR_0_3



MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 5: Sentiment alone (no surprise control)

The REG Procedure
Model: MODEL1
Dependent Variable: CAR_0_3



MODULE 07: Master Regression — Sentiment -> CAR
ROBUSTNESS 5: Sentiment alone (no surprise control)

The REG Procedure
Model: MODEL1
Dependent Variable: CAR_0_3

